

Code and Data Documentation for “Identifying Optimal Schistosomiasis Treatment Sequences for Mass Drug Administration Using Direct and Indirect Effects in Spatiotemporal Trials”

To reproduce all results presented in the paper, run `./Main.R`. Depending on the computing environment, the script may take a considerable amount of time to complete.

The following sections summarize the data and code used in the repository.

A. Data

The `./data` directory contains the following files.

File	Main objects	Notes
<code>Kenya_Score_Data_r.RData</code>	<code>Kenya_Score_Data</code> , <code>Site</code> , <code>Kenya.Dist.c</code> , <code>G.mat</code>	Kenya SCORE trial data, site-level information, and distance/adjacency matrices
<code>Tanzania_Score_Data_r.RData</code>	<code>Tanzania_Score_Data</code> , <code>Site</code> , <code>Tanzania.Dist.c</code> , <code>G.mat</code>	Tanzania SCORE trial data, site-level information, and distance/adjacency matrices
<code>Google_Kenya_Tanzania_Map.RData</code>	<code>ken.map</code> , <code>tan.map</code>	Google map objects used for geographic plots
<code>sim.Cov.Data.RData</code>	<code>sim.Cov.Data</code>	Simulation Scenario 1 and 3 covariance object; stored as a list of 5 regional sub-objects
<code>smoothed_surface.RData</code>	<code>simData.DataBase</code>	Simulation Scenario 2 smoothed surface object; stored as a list of 5 regional sub-objects

A.1 Using `Tanzania_Score_Data_r.RData` as an illustration

Objects in the file

Object	Class	Description
<code>Tanzania_Score_Data</code>	<code>data.table</code> , <code>data.frame</code>	Main Tanzania SCORE longitudinal trial data
<code>Site</code>	<code>data.frame</code>	Village/site summary information for Tanzania
<code>Tanzania.Dist.c</code>	<code>matrix</code> , <code>array</code>	Pairwise village distance matrix
<code>G.mat</code>	<code>matrix</code> , <code>array</code>	Adjacency matrix generated during preprocessing

Data frames and variables `Tanzania_Score_Data` (745 x 42)

#	Variables	Description
1	<code>Village_ID</code>	Village identifier
2	<code>Year</code>	Observation year
3	<code>Prevalence</code>	Observed infection prevalence
4	<code>Study_Type</code>	Study or trial type label

5	Study_Arm	Assigned treatment arm
6	Latitude	Village latitude
7	Longitude	Village longitude
8	flag	Region or geographic grouping label
9	CWT_1	Indicator for CWT at time lag 1
10	CWT_2	Indicator for CWT at time lag 2
11	CWT_3	Indicator for CWT at time lag 3
12	CWT_4	Indicator for CWT at time lag 4
13	SBT_1	Indicator for SBT at time lag 1
14	SBT_2	Indicator for SBT at time lag 2
15	SBT_3	Indicator for SBT at time lag 3
16	SBT_4	Indicator for SBT at time lag 4
17	CWT	Current CWT indicator
18	SBT	Current SBT indicator
19	IEt.CWT	Pow decay temporal variable for CWT
20	IEt.SBT	Pow decay temporal variable for SBT
21	IEt.CWT.exp	Exponential-decay temporal variable for CWT
22	IEt.SBT.exp	Exponential-decay temporal variable for SBT
23	sCWT_1	Smoothed or scaled CWT summary at lag 1
24	sCWT_2	Smoothed or scaled CWT summary at lag 2
25	sCWT_3	Smoothed or scaled CWT summary at lag 3
26	sSBT_1	Smoothed or scaled SBT summary at lag 1
27	sSBT_2	Smoothed or scaled SBT summary at lag 2
28	sSBT_3	Smoothed or scaled SBT summary at lag 3
29	IEs.No_Treatment	Spatial indirect-effect summary under no treatment
30	IEs.CWT.sp.Neigh.500.10	Spatial IE for CWT using 10-km as a range and a 500-km cut-off
31	IEs.SBT.sp.Neigh.500.10	Spatial IE for SBT using 10-km as a range and a 500-km cut-off
32	IEs.No_Treatment.sp.Neigh.500.10	Spatial IE for no treatment using 10-km as a range and a 500-km cut-off
33	IEs.CWT.sp.Neigh.500.30	Spatial IE for CWT using 30-km as a range and a 500-km cut-off
34	IEs.SBT.sp.Neigh.500.30	Spatial IE for SBT using 30-km as a range and a 500-km cut-off
35	IEs.No_Treatment.sp.Neigh.500.30	Spatial IE for no treatment using 30-km as a range and a 500-km cut-off
36	IEs.CWT.sp.Neigh.500.50	Spatial IE for CWT using 50-km as a range and a 500-km cut-off
37	IEs.SBT.sp.Neigh.500.50	Spatial IE for SBT using 50-km as a range and a 500-km cut-off
38	IEs.No_Treatment.sp.Neigh.500.50	Spatial IE for no treatment using 50-km as a range and a 500-km cut-off
39	distances	Distance to Lake Victoria
40	log.log.mean.Intensity	Mean of log-log transformed infection intensity
41	log.log.var.Intensity	Variance of log-log transformed infection intensity
42	Intercept	Intercept term used in model construction

Site (149 x 8)

#	Variables	Description
1	Village_ID	Village identifier
2	Study_Arm	Assigned treatment arm
3	LAT	Latitude
4	LON	Longitude
5	flag	Region or geographic grouping label
6	LON_X	Projected longitude coordinate
7	LAT_Y	Projected latitude coordinate
8	distances	Distance to Lake Victoria

B. Code for JSTVC

The source code for the JSTVC method is located in the `./JSTVC/` directory.

C. Simulation

All simulation code is located in the `./R/Simulation/` directory.

1. The `./R/Simulation/RandomField/` directory contains code for the first scenario using the Gneiting spatiotemporal covariance structure.
2. The `./R/Simulation/SmoothedFun/` directory contains code for the second scenario using low-rank basis expansion.
3. The `./R/Simulation/RandomField_xMisspecified_Decay/` directory contains code for the third scenario involving misspecification of the decay function.

D. Real Data Analysis

All real data analysis code is located in the `./R/Case/` directory.

E. Tables and Figures

1. Code for reproducing all tables is located in the `./table/` directory.
2. Code for reproducing figures from the simulation studies is located in the `./plot/simulation/` directory.
3. Code for reproducing figures from the exploratory data analysis (EDA) of the SCORE data is located in the `./plot/case/EDA/` directory.
4. Code for reproducing figures from the cross-validation analysis of the SCORE data is located in the `./plot/case/CV/` directory.
5. Code for reproducing figures from model fitting to the SCORE data is located in the `./plot/case/Fitting/` directory.
6. Code for reproducing figures related to treatment-sequence ranking for the SCORE data is located in the `./plot/case/Ranking/` directory.

F. Results

1. All figures are saved to the `./figure/` directory
2. All tables are saved to the `./result/summary` directory